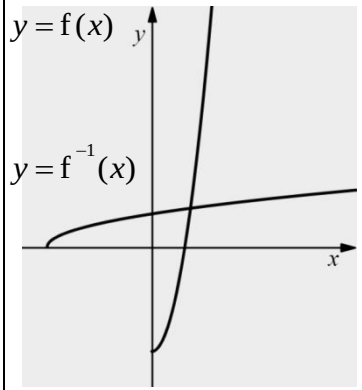
1. (a)	$g(3) = -265, \quad g(4) = 3104$ States change of sign, continuous and hence root in $[3, 4]$	M1 A1 **(2)**
(b)	$x_2 = \sqrt[6]{1000 - 2 \times 3} \qquad = 3.1591$ $(\alpha =) 3.1589$	M1 A1 A1 **(3)** **(5 marks)**

2 (a) (i)	$\log_6 T = 4 - 2\log_6 x$	B1
(ii)	E.g. $\log_6 T = 4 - 2\log_6 216 \Rightarrow \log_6 T = 4 - 2\times 3 = -2 \Rightarrow T = ...$	M1
	$\Rightarrow T = 6^{-2} = \frac{1}{36}$	A1
		(3)
(b)	$\log_6 T = 4 - 2\log_6 x \Rightarrow T = 6^{4-2\log_6 x}$	M1
	$\Rightarrow T = 6^4 \times 6^{\log_6 x^{-2}}$	dM1
	$\Rightarrow T = \frac{1296}{x^2}$	A1
		(3)
		(6 marks)

3 (i)	$\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(\sin^2 3x\right)=\frac{1}{\sin\cancel{3x}}\times 2\cancel{\sin 3x}\times 3\cos 3x=6\cot 3x$	M1 A1 **(2)**
(ii) (a)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(3x^2-4\right)^6=36x\left(3x^2-4\right)^5$	M1 A1 **(2)**
(b)	$\int x\left(3x^2-4\right)^5\mathrm{d}x=\frac{1}{36}\left(3x^2-4\right)^6$	B1ft
	$\int_0^{\sqrt{2}} x\left(3x^2-4\right)^5\mathrm{d}x=\left[\frac{1}{36}\left(3x^2-4\right)^6\right]_0^{\sqrt{2}}=\frac{1}{36}(2)^6-\frac{1}{36}(-4)^6=-112$	M1 A1cso **(3)** **(7 marks)**

4.(a)	$f \geqslant -5$	B1 **(1)**
(b)	 Curve starting on negative x-axis and passing through positive y-axis, in quadrants 1 and 2 only. Shape and position correct.	M1 A1 **(2)**
(c)	$2x^2 - 5 = x$ or $2x^2 - 5 = \sqrt{\frac{x+5}{2}}$ or $x = \sqrt{\frac{x+5}{2}}$ or $2\left(2x^2 - 5\right)^2 - 5 = x$	B1
	Full attempt to solve $2x^2 - x - 5 = 0 \Rightarrow x = ...$ exact	M1
	$x = \frac{1+\sqrt{41}}{4}$	A1 **(3)** **6 marks**

5 (i)	States $x = 2$	B1
	$\sqrt{3}\sec x + 2 = 0 \Rightarrow \cos x = -\dfrac{\sqrt{3}}{2} \Rightarrow x = ...$	M1
	$x = \dfrac{5\pi}{6}$	A1
		(3)
(ii)	Attempts to use $\cos 2\theta = 1 - 2\sin^2\theta$	M1
	$6\sin^2\theta + 10\sin\theta - 3 = 0$	A1
	$\sin\theta = \dfrac{-5 \pm \sqrt{43}}{6} \left(= -1.926..., 0.2595...\right) \Rightarrow \theta = \arcsin(...)$	M1
	$\theta = 15.0°,\ 165°$	A1
		(4)
		(7 marks)

6.(a)	$(2, -10)$	B1 B1 **(2)**
(b)	$ff(0) = f(-4) = ...$ $= 8$	M1 A1cso **(2)**
(c)	Attempts to solve $-3(x-2)-10 = 5x+10 \Rightarrow x = ...$ $x > -\frac{7}{4}$ only	M1 A1 **(2)**
(d)	$x\left(\text{or } \lvert x \rvert\right) = \frac{16}{3}$ Attempts $3(\lvert x \rvert - 2) - 10 = 0 \Rightarrow \lvert x \rvert = k,\ k > 0$ or $3(-x-2)-10 = 0 \Rightarrow x = -k$ or $3(x-2)-10 = 0 \Rightarrow x = k \Rightarrow x = -k$ $x = \left(\frac{16}{3} \text{ and }\right) -\frac{16}{3}$ with no other values	B1 M1 A1 **(3)** **(9 marks)**

7.(a)	States or implies that $A = 2\,500$	B1
	$10\,000 = 2500\mathrm{e}^{k\times 8} \Rightarrow 8k = \ln 4 \Rightarrow k = ...$	M1
	$\Rightarrow k = \frac{1}{8}\ln 4$ or awrt 0.1733	A1
		(3)
(b)	$\frac{\mathrm{d}N}{\mathrm{d}t} = 60\,000 \times -0.6\mathrm{e}^{-0.6\times 5} = -1792$ So decrease is 1790	M1, A1
		(2)
(c)	$60\,000\mathrm{e}^{-0.6t} = 2500\mathrm{e}^{0.1733t}$	M1
	$24 = \mathrm{e}^{0.1733t+0.6t} \Rightarrow 0.1733t + 0.6t = \ln 24 \Rightarrow t = ...$	dM1
	$T = 4.11$	A1
		(3)
		8 marks

8. (a)	$f'(x) = 6(2x+1)^2 e^{-4x} - 4(2x+1)^3 e^{-4x}$	M1 A1
	$= 2(2x+1)^2 e^{-4x}\{3-2(2x+1)\}$	dM1
	$= 2(2x+1)^2(1-4x)e^{-4x}$	A1
		(4)
(b)	Sets $f'(x) = 0 \Rightarrow x = -\frac{1}{2}, \frac{1}{4}$	B1
	Either $f\left("-\frac{1}{2}"\right) = \ldots$ or $f\left("\frac{1}{4}"\right) = \ldots$	M1
	Both $\left(-\frac{1}{2}, 0\right)$ and $\left(\frac{1}{4}, \frac{27}{8e}\right)$	A1
		(3)
(c)	$\left(\frac{9}{4}, \frac{27}{e}\right)$	B1ft B1ft
		(2)
		9 marks

9(a)	$\frac{\cos 2x}{\sin x}+\frac{\sin 2x}{\cos x}=\frac{\cos 2x}{\sin x}+\frac{2\sin x\cos x}{\cos x}$ (One Correct identity)	B1
	$=\frac{1-2\sin^2 x}{\sin x}+\frac{2\sin x\cancel{\cos x}}{\cancel{\cos x}}$	M1
	$=\frac{1}{\sin x}-\frac{2\sin\cancel{^2}x}{\cancel{\sin x}}+2\sin x=\frac{1}{\sin x}=\text{cosec } x$ *	A1*
		(3)
(b)	E.g. Equation is $\text{cosec}^2\theta=6\cot\theta-4\Rightarrow 1+\cot^2\theta=6\cot\theta-4$	M1
	E.g. $\cot^2\theta-6\cot\theta+5=0$	A1
	E.g. $\tan\theta=\frac{1}{5},1$	dM1
	$\theta=0.197,\frac{\pi}{4}$	A1, A1
		(5)
(c)	$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\text{cosec}x\cot x\,\text{d}x=\left[-\text{cosec}x\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$	M1
	$=2-\sqrt{2}$	A1
		(2)
		10 marks

(a) **ALT I**	$\dfrac{\cos 2x}{\sin x}+\dfrac{\sin 2x}{\cos x}=\dfrac{\cos 2x\cos x+\sin 2x\sin x}{\sin x\cos x}$	Correct single fraction	B1
	$=\dfrac{\cos x\left(1-2\sin^2 x\right)+2\sin x\cos x\sin x}{\sin x\cos x}$	Single fraction with single arguments and common factor $\cos x$ in numerator	M1
	$=\dfrac{\cancel{\cos x}}{\sin x\cancel{\cos x}}=\dfrac{1}{\sin x}=\text{cosec } x$ *		A1* **(3)**
(a) **ALT II**	$\dfrac{\cos 2x}{\sin x}+\dfrac{\sin 2x}{\cos x}\equiv\dfrac{\cos 2x\cos x+\sin 2x\sin x}{\sin x\cos x}$	Correct single fraction	B1
	$\equiv\dfrac{\cos\left(2x-x\right)}{\sin x\cos x}$	Applies identity to reach single fraction with single arguments and common factor $\cos x$ in numerator	M1
	$\equiv\dfrac{\cancel{\cos x}}{\sin x\cancel{\cos x}}\equiv\dfrac{1}{\sin x}\equiv\text{cosec } x$ *		A1* **(3)**
(a) **ALT III**	$\dfrac{\cos 2x}{\sin x}+\dfrac{\sin 2x}{\cos x}\equiv\dfrac{\cos^2 x-\sin^2 x}{\sin x}+\dfrac{2\sin x\cos x}{\cos x}$	Correct identity	B1
	$\equiv\dfrac{\cos^3 x-\sin^2 x\cos x+2\sin^2 x\cos x}{\sin x\cos x}$	Single fraction with single arguments and common factor $\cos x$ in numerator	M1
	$\equiv\dfrac{(\cos^2 x+\sin^2 x)\cancel{\cos x}}{\sin x\cancel{\cos x}}\equiv\dfrac{1}{\sin x}\equiv\text{cosec } x$ *		A1* **(3)**

10 (a)	$x=\dfrac{2y^2+6}{3y-3} \Rightarrow \left(\dfrac{\mathrm{d}x}{\mathrm{d}y}=\right)\dfrac{4y(3y-3)-3(2y^2+6)}{(3y-3)^2}$	M1 A1
	$\dfrac{\mathrm{d}x}{\mathrm{d}y}=\dfrac{6y^2-12y-18}{9(y-1)^2}=\dfrac{2y^2-4y-6}{3(y-1)^2}$ o.e	dM1, A1
		(4)
(b)	P and Q are where $\dfrac{\mathrm{d}x}{\mathrm{d}y}=0$ or where $2y^2-4y-6=0$	B1
	Solves $2y^2-4y-6=0 \Rightarrow 2(y-3)(y+1)=0 \Rightarrow y=3,-1$	M1
	Subs $y=-1$ and 3 in $x=\dfrac{2y^2+6}{3y-3} \Rightarrow x=..$	dM1
	Achieves $x=-\dfrac{4}{3}$ and $x=4$	A1cso
		(4)
		8 marks

Alt (a)	$x=\dfrac{2y^2+6}{3y-3}\Rightarrow 3xy-3x=2y^2+6\Rightarrow 3x+3y\dfrac{\mathrm{d}x}{\mathrm{d}y}-3\dfrac{\mathrm{d}x}{\mathrm{d}y}=4y$ $\dfrac{\mathrm{d}x}{\mathrm{d}y}=\dfrac{4y-3x}{3(y-1)}$	M1 A1 dM1, A1 **(4)**
(b) First 2 marks.	States that P and Q are where $\dfrac{\mathrm{d}x}{\mathrm{d}y}=0$ or where $4y-3x=0$ $\Rightarrow \dfrac{4}{3}y=\dfrac{2y^2+6}{3y-3}\Rightarrow 4y^2-4y=2y^2+6\Rightarrow$ as main scheme	B1 M1
Alt II (a)	$x=\dfrac{2y^2+6}{3y-3}=\dfrac{2y}{3}+\dfrac{2}{3}+\dfrac{8}{3(y-1)}\Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}y}=\dfrac{2}{3}-\dfrac{8}{3(y-1)^2}$ $\dfrac{\mathrm{d}x}{\mathrm{d}y}=\dfrac{2(y-1)^2-8}{3(y-1)^2}=\dfrac{2y^2-4y-6}{3(y-1)^2}$ oe	M1 A1 dM1, A1 **(4)**